

How can we use Active Learning Strategies to promote **Metacognition**?

Metacognition: "Thinking about your own thinking" or in a classroom setting, we can think of it as helping students to think about ***What they know about what they know*** and avoid ***The Illusion of Mastery***.

How can we help? One idea is to make frequent use of **testing and retrieval practice**.

Testing: Provide frequent low-stakes self-assessments with immediate feedback. Studies have shown that negative feedback is more helpful for students when distinguishing what they actually know. The feedback should be corrective, and elaborative. (Give the **WHY**)

Retrieval Practice: Retrieval practice interrupts the process of forgetting. Retrieval practice is more about remembering facts than practicing skills. We want them to pull from their memory. Simple and fast activities like "brain dumps" from the pre-class assignments can be done in just a few minutes.

(A1): Work with a partner. One of you will write down what steps you would take to solve the following question, and the other will solve the problem. Change your role on #2.

4.1 Self-Check Exercises

1. Find the intervals where the function

$$f(x) = \frac{2}{3}x^3 - x^2 - 12x + 3$$

is increasing and the intervals where it is decreasing.

Answer ↓

When you have completed both parts, compare with your partner.

2. Now find the intervals on which $f(x)$ is concave up and concave down. Provide the x coordinate of inflection point(s) if applicable.

Check your work:

1. Find the intervals where the function

$$f(x) = \frac{2}{3}x^3 - x^2 - 12x + 3$$

is increasing and the intervals where it is decreasing.

Answer +

The derivative of f is

$$f'(x) = 2x^2 - 2x - 12 = 2(x + 2)(x - 3)$$

and it is continuous everywhere. The zeros of $f'(x)$ are $x = -2$ and $x = 3$. The sign diagram of f' is shown in the accompanying [figure](#). We conclude that f is increasing on the intervals $(-\infty, -2)$ and $(3, \infty)$ and decreasing on the interval $(-2, 3)$.



1. For Concavity, you will need to find $f''(x)$.

$$f''(x) = 4x - 2$$

2. Now, set this equal to zero and solve. Also, check where $f''(x)$ DNE, in this case nowhere.

3. Place $x = \frac{1}{2}$ on a number line. Plug values into $f''(x)$ to get the sign, + or -

4. Clearly state your intervals without including the endpoints.

5. Determine that $x = \frac{1}{2}$ is an inflection point.

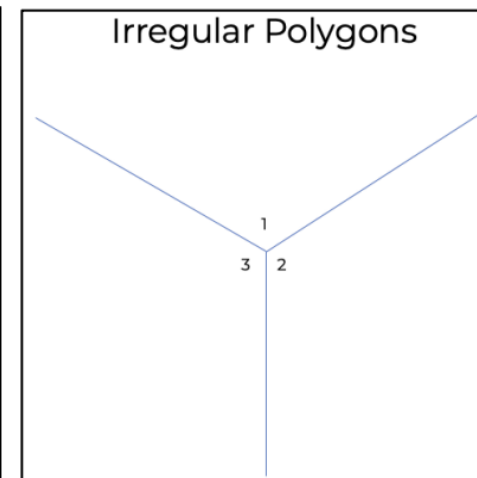
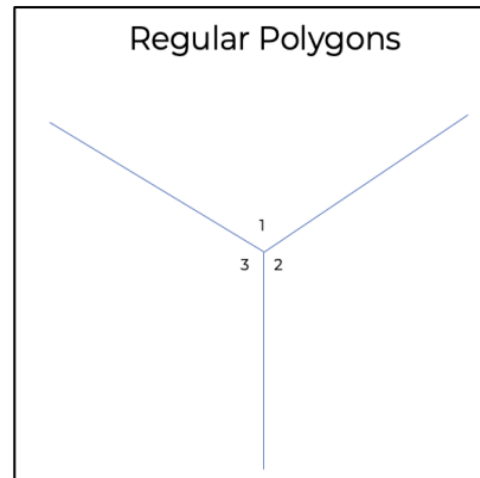
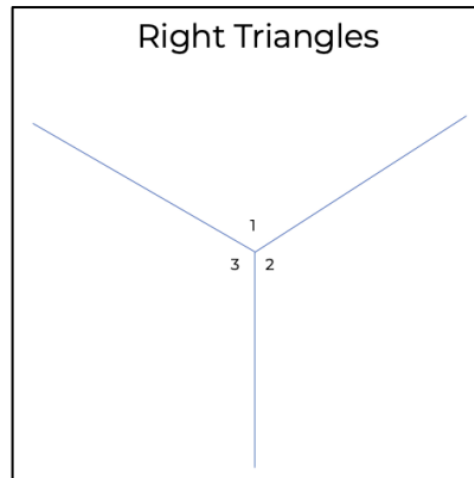
1. We need CN, so find $f'(x)$.
2. Set $f'(x) = 0$ and solve for x . Don't forget to check where $f'(x)$ DNE! In this case, there are no values
3. Place these on a number line. Choose a value to the left and right to plug into $f'(x)$. Check for + or -
4. Provide the intervals where $f(x)$ is increasing or decreasing. DO NOT include the endpoints. **WHY?**

What have you learned? Write down what questions you have (and then get answers!) and give yourself some feedback on how well you know A1.

In addition to simple activities like Brain Dumps, or having each table ask a question, here is another retrieval activity. I use blank pages from my extra lesson copies. In pairs, divide the sheet into two columns. Choose two ideas, which can be as simple as domain and range. This term I used it with AROC and IROC.

8. Pass the Paper

Students work in groups of three. Each student has a piece of paper, divided into three sections. There is a different topic listed on the top of each paper. Students are given 2-5 minutes to record everything they know about that topic in Section 1. On your cue, students rotate papers (and thus topics), review what previous group members contributed, and add anything else they know to Section 2 of that paper. The process is repeated a third time so each student is contributing to each paper/topic.



Takeaways and Points for Discussion

- What do we already do? KC reflections/ practice exams. But the efficacy of each depends on **feedback**. Ideally, we provide both elaborative (explaining the why) and corrective. If a student does not receive timely feedback or use the practice exams correctly, these do not achieve the desired impact. Providing opportunities where we can ensure they get the feedback is more helpful than depending on them.
- The Dunning-Krueger Effect: A cognitive bias where an individual lacks the skills to improve because they are unable to distinguish between incompetence and competence. One of the reasons why, is a lack of feedback, particularly negative feedback. Even after training, the least competent students still overestimate what they know.
- Note: Equally unhelpful for our students is the "curse of knowledge" that many instructors suffer from. Our tendency to underestimate how long it will take another person to learn something new or perform a task that we have mastered. Partially because as you know more about something, your models grow more complex and the component steps fade into the background. Think about how an elite swimmer doesn't think about all the "components" of a good backstroke after it is mastered. But if he were to teach someone the backstroke, it may start with a simple key component of looking straight up!